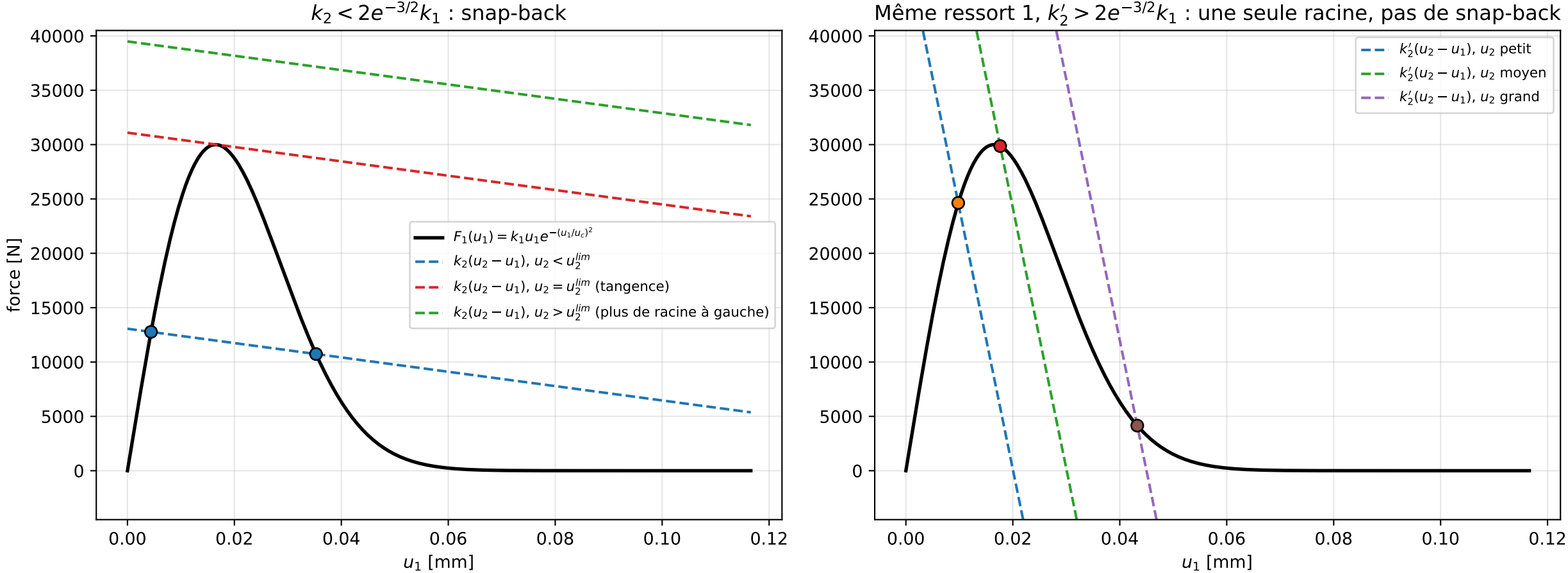
$k_2 < 2e^{-3/2}k_1$: snap-back
force [N]
u_1 [mm]
$F_1(u_1) = k_1 u_1 e^{-(u_1/u_c)^2}$
$k_2(u_2 - u_1)$, $u_2 < u_2^{lim}$
$k_2(u_2 - u_1)$, $u_2 = u_2^{lim}$ (tangence)
$k_2(u_2 - u_1)$, $u_2 > u_2^{lim}$ (plus de racine à gauche)
Même ressort 1, $k_2' > 2e^{-3/2}k_1$: une seule racine, pas de snap-back
u_1 [mm]
$k_2'(u_2 - u_1)$, u_2 petit
$k_2'(u_2 - u_1)$, u_2 moyen
$k_2'(u_2 - u_1)$, u_2 grand